



Figure 0: Boeing 737 MAX

Mech 460 Case Study - Boeing 737 MAX
Quantitative Analysis Report
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Due: 4/14/2025

Introduction

The Boeing 737 MAX introduced significant changes over the 737 Next Generation (NG), including larger, more efficient engines and aerodynamic improvements. These design shifts altered weight distribution, propulsion performance, and longitudinal stability. Boeing addressed pitch stability challenges using a software solution called the Maneuvering Characteristics Augmentation System (MCAS) instead of a major structural redesign. This report compares the NG and MAX across three key areas:

1. Longitudinal static stability and CG location
2. Propulsion efficiency and fuel burn
3. The technical and financial rationale for choosing MCAS over hardware redesign

1. Center of Gravity and Static Stability Analysis

1.1 CG Shifts Due to Engine and Gear Changes

The 737 MAX 8 and 737-800 NG have nearly identical fuselage lengths (~39.5 m), but the MAX uses CFM LEAP-1B engines, which are ~1,800 lb heavier and mounted ~1 ft farther forward for ground clearance. This shifts the CG slightly forward, although the change remains small relative to the aircraft's chord line. The following equation finds the center of gravity by adding all the moments (weight * position from neutral axis) and dividing by the total weight.



Figure 1: Repositioned Engine

Center of Gravity equation:

$$X_{cg} = \frac{\sum W_i \cdot X_i}{\sum W_i}$$

W_i = Component weights (engines, fuselage, tail, etc.)

X_i = Longitudinal positions of each component

Component	737-800 NG (CFM56-7B)	737 MAX 8 (LEAP-1B)	Notes
Engine	~4,300 lb	~6,100 lb at X+Δ	Δ ≈ 1 ft
Wing + Gear	~20,000 lb	~21,000 lb	Strengthened wing; 8 in longer nose gear
Fuselage Systems	~50,000 lb	~53,000 lb	Heavier avionics/systems
Total OEW	~91,300 lb	~98,500 lb	~7,200 lb increase

Table 1 – Key Weight Differences

1.2 Static Margin and Pitch Stability

Despite a modest CG shift, the MAX faced pitch-up tendencies at high angles of attack (AoA) due to engine nacelle lift. The static margin quantifies pitch stability. The static margin represents the distance between the CG and the neutral point, expressed as a percentage of the wing's mean aerodynamic chord (MAC). A positive static margin means the aircraft naturally resists pitching up too far. A reduced static margin means it's easier for the aircraft to unintentionally keep pitching up. The formula for the static margin follows:

$$\text{Static Margin} = \frac{(X_{np} - X_{cg})}{c_{MAC}}$$

X_{np} = neutral point

X_{cg} = center of gravity

c_{MAC} = mean aerodynamic chord

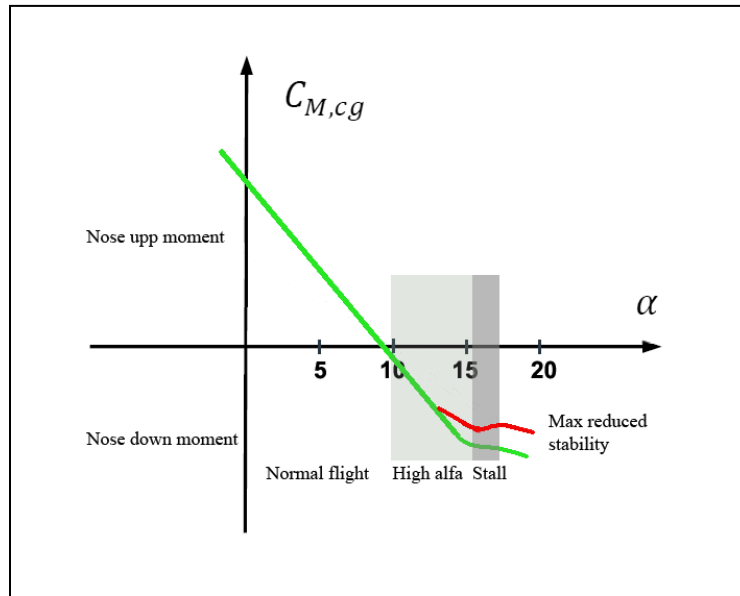


Figure 2: Pitch Moment Curve of 737 Max

To illustrate how pitch moment changes with CG, we calculate the pitch moment coefficient:

$$C_m = C_{m,ac} + \left[\frac{(X_{cg} - X_{ac})}{c} \right] \cdot CL + CL_t \cdot \left(\frac{S_t}{S} \right) \cdot \frac{l_t}{c}$$

$C_{m,ac}$ = wing-body moment at aerodynamic center

X_{cg} = center of gravity

CL = main wing lift coefficient

CL_t = tail lift coefficient

$\frac{S_t}{S}$ = tail area ratio

$\frac{l_t}{c}$ = moment arm ratio

Because the forward engine lift reduced pitch stability, Boeing had to find a solution to ensure the aircraft still behaved safely and predictably, especially at high angles of attack. Rather than redesigning the airframe (which would be expensive), Boeing introduced MCAS, a software-based fix that automatically pushes the nose down in certain conditions to restore stability and mimic the handling of earlier 737 models. This decision, while cost-effective, introduced a new failure risk, highlighted by the accidents that followed.

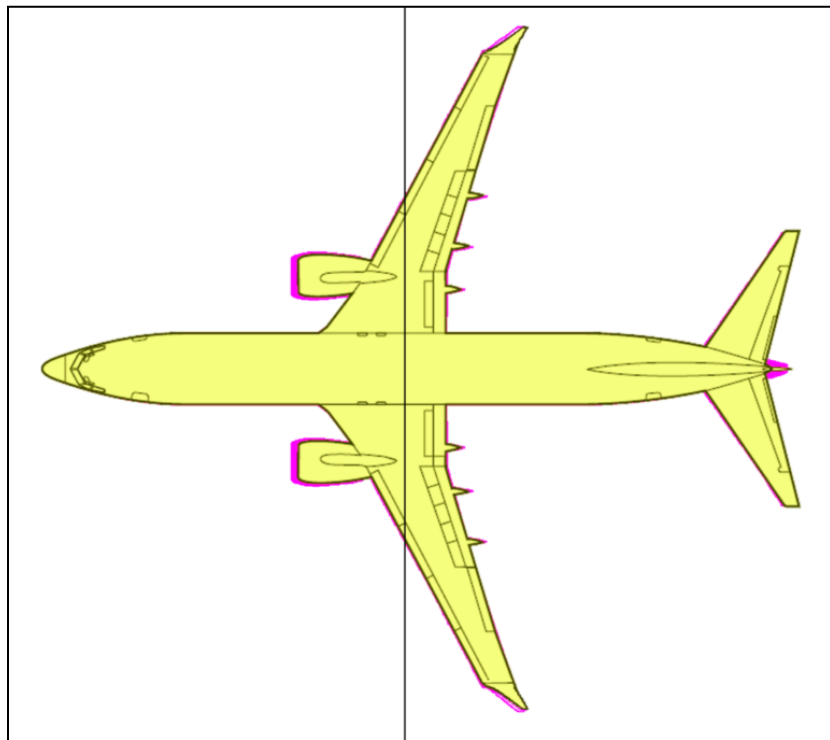


Figure 3: Geometry of 737 NG on top of MAX

2. Engine Performance and Fuel Efficiency Comparison

2.1 LEAP-1B Engine vs. CFM56-7B Engine– Specifications

The LEAP-1B improves upon the CFM56-7B with better thermal and propulsive efficiency. Key specs:

Metric	CFM56-7B (737 NG)	LEAP-1B (737 MAX)
Thrust (max)	~27,300 lbf	~28,000 lbf
Bypass Ratio	~5.1:1	~8.6:1
OPR	~32–33:1	~40–50:1
Fan Diameter	~61 in	69.4 in
TSFC (cruise est.)	~0.60 lb/(lbf·hr)	~0.51 lb/(lbf·hr)

Table 2 – Specification Comparison

The LEAP-1B provides ~15% better TSFC (fuel burn per unit thrust per hour), which is essential for competing with the Airbus A320neo.

2.2 Breguet Range and Fuel Burn Estimates

Range formula:

$$R = \left(\frac{V}{c}\right) \cdot \frac{L}{D} \cdot \log\left(\frac{W_i}{W_f}\right)$$

V = cruise velocity

c = TSFC

$\frac{L}{D}$ = lift-to-drag ratio

$\frac{W_i}{W_f}$ = initial/final weight

With a 15% lower TSFC and slight improvements in L/D from aerodynamic tweaks, the MAX achieves ~14–20% greater range or equivalent fuel savings. Boeing claims up to \$1M annual savings per jet over typical flight cycles—equaling \$15–20M over a 20-year service life (baatraining.com).

3. MCAS vs Hardware Redesign: Technical and Financial Trade-offs

3.1 Engineering Rationale

Installing larger engines led to high-AoA instability. Boeing could have pursued:

- Hardware Fix: new wing, larger horizontal tail, taller gear
- Software Fix: MCAS trims nose-down automatically at high AoA

Hardware fixes are heavy, expensive, and delay certification. A new wing or gear redesign could cost \$1–2B and push recertification timelines years into the future. To preserve the 737 type rating and pilot commonality, Boeing embedded MCAS into the existing speed trim system.

3.2 Cost Comparison

Cost Category	Hardware Redesign	MCAS Approach
Development Cost	\$1–2 billion (R&D)	~\$200 million (software)
Recurring Cost	Higher manufacturing cost	Minimal
Added Weight	Yes – reduces fuel savings	None
Certification Delay	2–4 years	Minimal
Training Cost	Simulator required	Avoided pre-crisis

Table 3 – Cost Comparison

To fight pitch instability from the larger LEAP-1B engines at high angles of attack, Boeing arrogantly chose a software solution, MCAS, over more costly hardware redesigns. While structural fixes like a new wing, extended landing gear, or a larger tail would have improved stability, they would have added weight, cost \$1–2 billion, delayed certification by years, and required simulator training for pilots. In contrast, MCAS cost an estimated \$200 million to develop, added no weight, and allowed Boeing to maintain the 737 type rating and avoid expensive requirements. MCAS was integrated into the speed trim system to act only during high-AoA conditions, trimming the stabilizer nose-down automatically. However, it relied on a single angle-of-attack (AoA) sensor, introducing a critical failure point. The lack of redundancy and limited pilot awareness caused two crashes, ultimately costing Boeing an estimated \$20 billion in legal, compensation, and order-related losses. In order for future Boeing 737 Max model's certification, it was made necessary by regulatory bodies to have at least 3 AoA sensors on any aircrafts relying on MCAS as a pitch stabilizer.

4. Conclusions

The 737 MAX was designed for efficiency and competitiveness. The LEAP-1B engine and aerodynamic changes enabled 14–15% fuel savings over the 737 NG, fulfilling market needs. However, the decision to handle stability concerns via MCAS instead of redesigning hardware, highlighted the dangers of relying solely on software for flight-critical functions without redundancy or training transparency. Although the CG shift from the new engines was small, their forward position introduced destabilizing aerodynamic effects at high angles of attack, especially during takeoff or go-around. MCAS was implemented to counteract these effects by trimming nose-down automatically, but it depended on a single angle-of-attack sensor and lacked pilot awareness or sufficient manual override training. Although MCAS preserved certification simplicity and reduced costs short-term, it introduced system-level risks that had catastrophic consequences.

In addition, these poor bureaucratic decisions caused 346 lives to be lost. The 737 MAX case serves as a sobering lesson in aircraft design: when software compensates for physical limitations, it must be engineered and communicated with exceptional care. Redundancy, pilot preparedness, and transparency must never be sacrificed for market speed or cost savings.

Appendix: Equations

Appendix a - Governing Equations

CG Location:

$$X_{cg} = \frac{\sum W_i \cdot X_i}{\sum W_i}$$

Static Margin:

$$Static\ Margin = \frac{(X_{np} - X_{cg})}{c_{MAC}}$$

Pitch Moment Coefficient:

$$C_m = C_{m,ac} + \left[\frac{(X_{cg} - X_{ac})}{c} \right] \cdot CL + CL_t \cdot \left(\frac{S_t}{S} \right) \cdot \frac{I_t}{c}$$

Breguet Range Equation:

$$R = \left(\frac{V}{c} \right) \cdot \frac{L}{D} \cdot \log\left(\frac{W_i}{W_f}\right)$$

Appendix b - References

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